

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	6

Case Study Topic: Steel Rod Cutting Profit Maximization using Dynamic Programming

Word Done Summary:

In this case study, we analysed the steel rod-cutting optimization problem using Dynamic Programming. Tata Steel's rod-cutting process was modelled to maximize total revenue by dividing rods into smaller pieces. We implemented bottom-up DP tabulation to compute optimal revenues and reconstructed the optimal cutting sequence. We also compared memorized top-down and bottom-up approaches along with different variants of the cutting-stock problem.

Implementation Details:

1. Represented rod prices using an integer array.
2. Applied Dynamic Programming tabulation.
3. Computed $dp[i]$ for rod lengths from 1 to 10.
4. Stored optimal cut positions.
5. Reconstructed the optimal cutting sequence.
6. Compared top-down memorization and bottom-up DP.
7. Studied variants of the cutting-stock optimization problem.

Result:

Dynamic Programming successfully computed the maximum obtainable revenue for rod cutting. The bottom-up DP approach efficiently solved the optimization problem while reconstructing the optimal cutting strategy.

Output:

```
DP Table:  
dp[0] = 0  
dp[1] = 1  
dp[2] = 5  
dp[3] = 8  
dp[4] = 10  
dp[5] = 13  
dp[6] = 17  
dp[7] = 18  
dp[8] = 22  
dp[9] = 25  
dp[10] = 30
```

Maximum Revenue: 30

GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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